Achieving low-temperature sintering of SiC ceramics (below 1450°C) requires a fundamental shift from conventional solid-state sintering, which typically demands temperatures above 1800°C due to the strong covalent Si–C bonds. The strategies that enable this reduction fall into three main categories: field-assisted sintering with specialized powder preparation, reaction bonding, and liquid-phase sintering with carefully selected additives.

Raw Materials for Low-Temperature Sintering

SiC Powders

The starting SiC powder characteristics are critical for low-temperature densification. **Granulated α-SiC powders** produced by spray-drying have proven particularly effective. These granules, typically tens of micrometers in size, are composed of fine α-SiC particles with an average size of ~100 nm and are held together by organic binders such as polyvinyl alcohol (PVA) and polyethylene glycol (PEG). The organic coating on these granules plays a dual role: it provides flowability for die filling and, during sintering, decomposes in situ to form intergranular graphite that enhances densification.

For liquid-phase sintering routes, β-SiC nanopowder has been used with Li_2O doping combined with Y_2O_3-Al_2O_3-MgO sintering aids. The nano-scale particle size increases the driving force for densification, while the oxide additives form a liquid phase that facilitates particle rearrangement and mass transport.

Sintering Aids for Liquid-Phase Routes

Several additive systems have demonstrated effectiveness at reducing SiC sintering temperatures to or near the 1450°C threshold:

Al_2O_3–CeO_2 system: The combination of Al_2O_3 and CeO_2 forms a eutectic liquid phase with good wettability on SiC surfaces. This system enables liquid-phase sintering at temperatures significantly lower than the conventional Al_2O_3–Y_2O_3 system, which typically requires temperatures above 1900°C despite having a eutectic temperature of 1760°C. CeO_2 is

particularly advantageous because it is a tetravalent rare earth oxide with abundant reserves and low cost, and it exhibits low reactivity with SiC, preventing decomposition during sintering.

Li_2O with Y_2O_3-Al_2O_3-MgO: This multi-component additive system has been used to achieve densification of β-SiC nanopowder at temperatures as low as 1450°C when combined with spark plasma sintering.

NaA zeolite residue: Industrial waste from zeolite membrane production, rich in Na, Al, and Si, has been used as a low-cost sintering aid for porous SiC ceramics. With this additive, SiC supports have been prepared at 1450°C with bending strength of 22.5 MPa. Other low-cost additives include CaO and ZrO_2, which have achieved similar results.

Reaction Bonding Raw Materials

For reaction-bonded SiC (RBSC), the raw materials include:

- **SiC powder** (particle sizes from 0.5 μm to several hundred micrometers)
- **Carbon sources**: carbon black (10–40 nm) or graphite, typically 0–30 vol%
- **Silicon metal powder** (70–2000 μm) as the infiltrant
- **Binders**: phenolic resin, PVA, or sodium lignosulfonate

The carbon source reacts with molten silicon to form new SiC, bonding the pre-existing SiC particles together. This reaction occurs at temperatures of **1410–1460°C** under vacuum, which falls within the target low-temperature range.

Sintering Methods and Mechanisms

Electric-Current Confinement Sintering (ECS)

This advanced field-assisted technique has achieved **96.39% relative density at 1435°C**, approximately 350°C lower than conventional spark plasma sintering. The method employs an insulating boron nitride (BN) die that confines the entire electric current flow to the specimen, dramatically increasing local current density. The key innovation lies in using granulated SiC powders with organic binders: during heating, the organic coating decomposes in situ to form **highly ordered intergranular graphite flakes**. These graphite flakes serve two functions: they provide solid lubrication for particle rearrangement during the early stages of densification, and they enhance fracture energy absorption through crack deflection in the final ceramic.

Spark Plasma Sintering (SPS) with Granulated Powders

Combining SPS with granulated powders enables additive-free densification of α-SiC at **1750°C**, with relative density exceeding 98%. While this is above 1450°C, it represents a 100–200°C reduction compared to conventional pure SiC sintering. The rapid heating capability of SPS effectively cleans and activates particle surfaces, promoting the formation of sintering necks. Granulated powders exhibit superior flowability, which benefits both die filling and densification. The resulting ceramics achieve Vickers hardness of 26.3 GPa, bending strength of 461 MPa, and fracture toughness of 4.6 $MPa \cdot m^{1/2}$.

Reaction Bonding

Reaction bonding is the most industrially established low-temperature approach for SiC. The process involves:

1. Forming a porous preform from SiC powder and a carbon-yielding binder
2. Heating the preform to convert the binder to carbon
3. Contacting the preform with a silicon source (silicon cake or granules)
4. Heating to **1410–1460°C** under vacuum, causing molten silicon to infiltrate and react with carbon to form new SiC

The reaction-bonded SiC achieves densities of **96.7–97.8%** depending on the carbon black content in the preform. The process is economically attractive because it requires only a reaction sintering furnace, and the exothermic reaction between silicon and carbon helps maintain

uniform temperature during infiltration. The resulting material is a composite of SiC (continuous phase) and residual silicon (discontinuous phase, <25%).

Liquid-Phase Sintering with Al_2O_3–CeO_2

Pressureless liquid-phase sintering with Al_2O_3–CeO_2 additives has been demonstrated as a suitable low-temperature route for α-SiC. The mechanism relies on three criteria being satisfied: (1) the liquid phase wets SiC particles, (2) SiC dissolves into the liquid phase, and (3) the liquid phase penetrates along SiC–SiC grain boundaries. While the specific densification temperature for this system is reported relative to the conventional Al_2O_3–Y_2O_3 system (which sinters above 1900°C), the Al_2O_3–CeO_2 eutectic enables processing at substantially lower temperatures, with mechanical properties (flexural strength, hardness, fracture toughness) comparable to the higher-temperature system.

Summary Table

Method	Raw Materials	Sintering Temperature	Achieved Density	Key Mechanism
ECS	Granulated α-SiC (PVA/PEG binder)	1435°C	96.4%	In-situ graphite from binder decomposition
SPS + Granulated	Granulated 6H-SiC (spray-dried)	1750°C	>98%	Surface activation + particle rearrangement
Reaction Bonding	SiC + carbon black + Si powder	1410–1460°C	96.7–97.8%	Molten Si infiltration + reaction
LPS (Al_2O_3–CeO_2)	α-SiC + Al_2O_3 + CeO_2	<1900°C (substantially reduced)	Dense	Eutectic liquid phase wetting/dissolution

The most promising route for achieving SiC densification below 1450°C is the combination of reaction bonding (1410–1460°C) and electric-current confinement sintering with granulated powders (1435°C). Both approaches leverage in-situ reactions or decomposition products to circumvent the kinetic limitations imposed by SiC's strong covalent bonding.